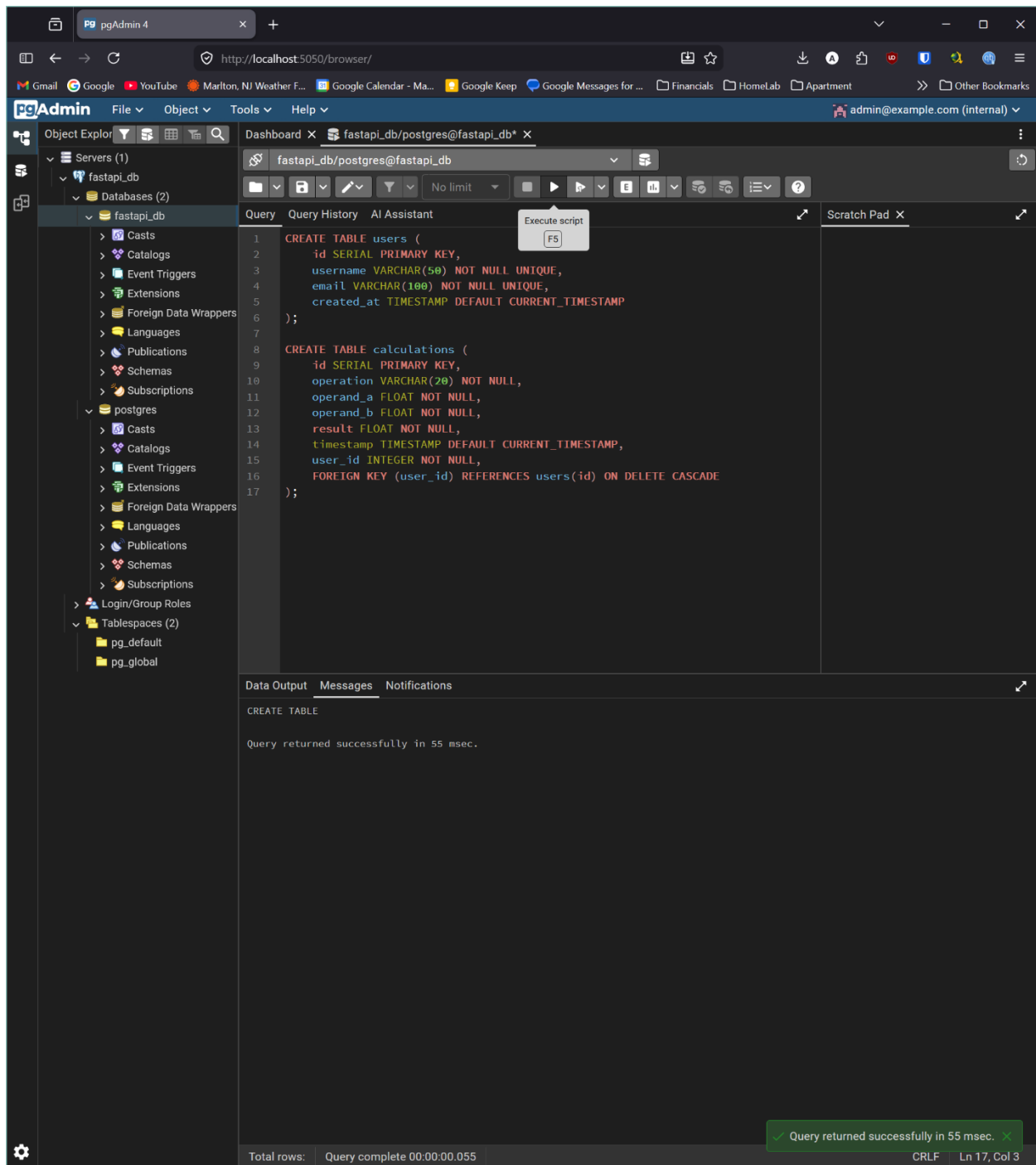
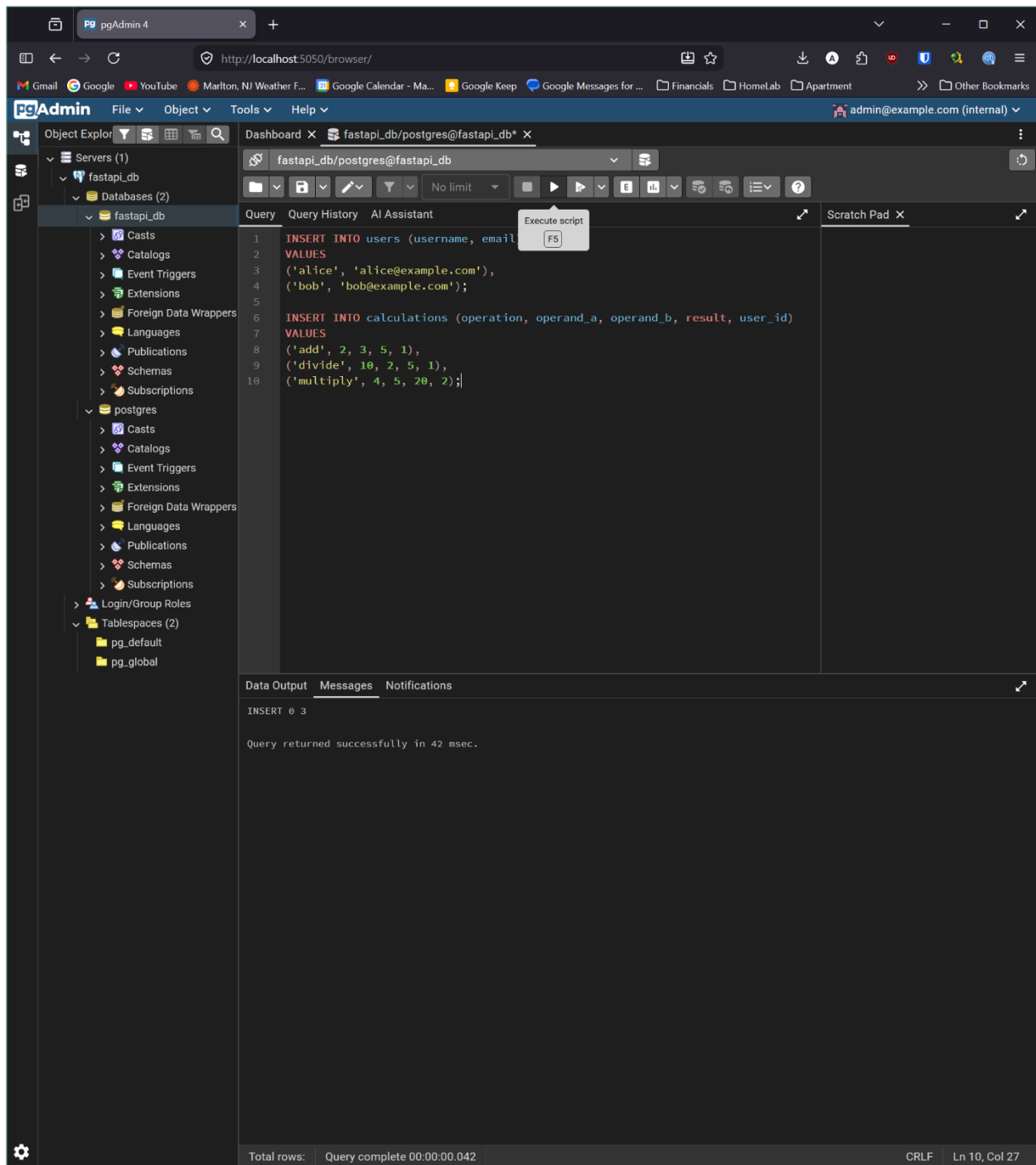




Two tables are created: users to store user accounts with a unique username and email, and calculations to store math operations linked to a user via a foreign key (user_id).



Sample data is inserted — two users (alice and bob) and three calculations (add and divide for alice, multiply for bob) — to populate the tables for testing.

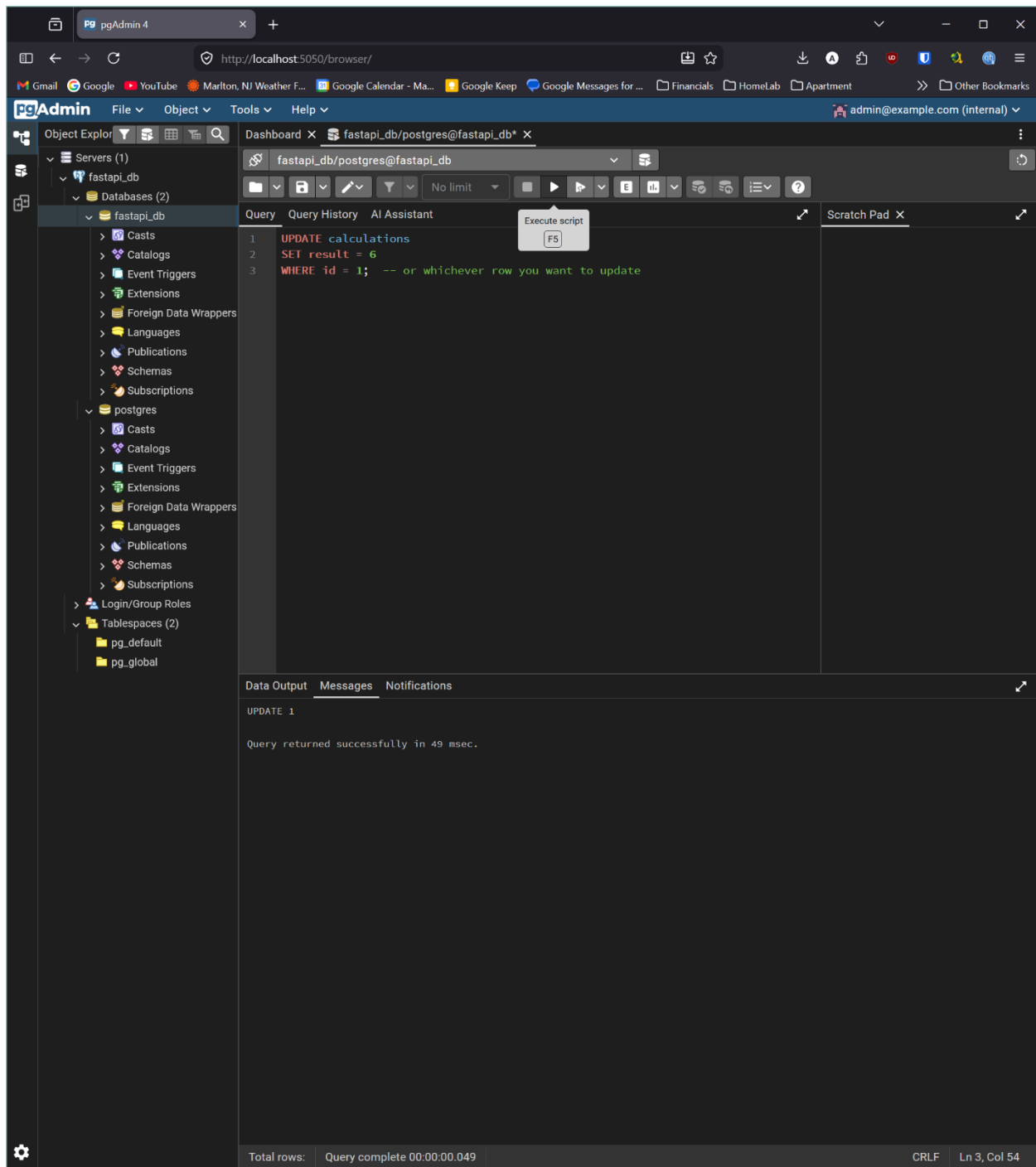
The screenshot shows the pgAdmin 4 web interface in a browser. The left sidebar displays the database structure for 'fastapi_db', including 'Databases (2)' with 'fastapi_db' and 'postgres'. The main panel shows a SQL query in the 'Query' tab. The query consists of three parts: a comment, a SELECT statement for users, another comment, a SELECT statement for calculations, and a JOIN query combining the two. An 'Execute script' button with a keyboard shortcut 'F5' is visible above the query. Below the query, the 'Data Output' tab shows the results of the JOIN query as a table with 3 rows. A status bar at the bottom indicates the query was successfully run with a runtime of 51 msec, affecting 3 rows.

```
1  -- Retrieve all users
2  SELECT * FROM users;
3
4  -- Retrieve all calculations
5  SELECT * FROM calculations;
6
7  -- Join users and calculations
8  SELECT u.username, c.operation, c.operand_a, c.operand_b, c.result
9  FROM calculations c
10 JOIN users u ON c.user_id = u.id;
```

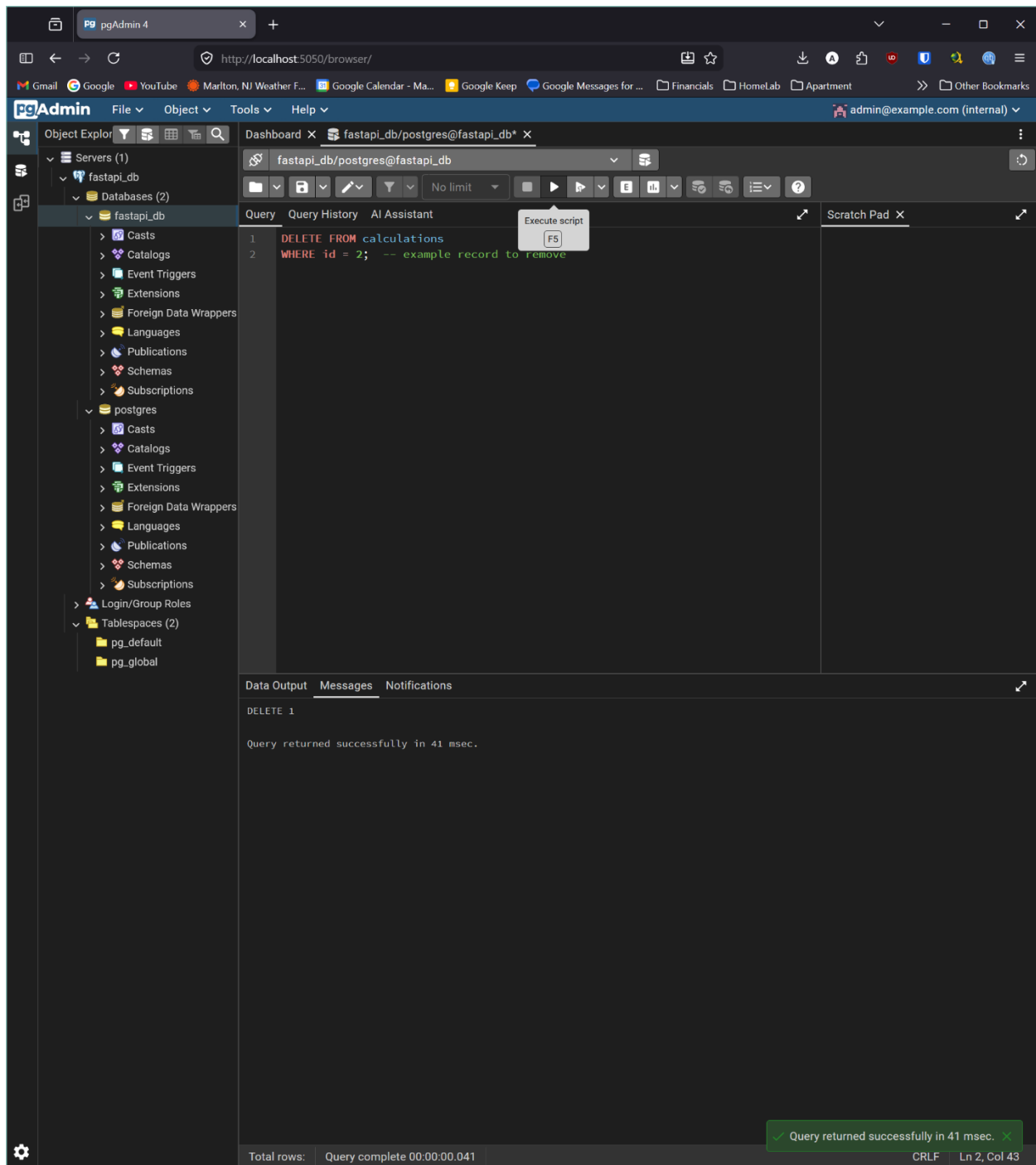
	username character varying (50)	operation character varying (20)	operand_a double precision	operand_b double precision	result double precision
1	alice	add	2	3	5
2	alice	divide	10	2	5
3	bob	multiply	4	5	20

Successfully run. Total query runtime: 51 msec. 3 rows affected.

Three SELECT queries retrieve the data: one for all users, one for all calculations, and a JOIN that combines both tables to show which user performed each calculation.



The result column of calculation id=1 is corrected from 5 to 6 using an UPDATE statement, demonstrating how to modify existing records.



Calculation id=2 is permanently removed from the table using a DELETE statement, demonstrating how to remove individual records without affecting the rest of the data.